Notes on Decomposition Spaces in Homotopy Combinatorics

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Abstract

This note is written after reading [2], [4], [5], and [1]. We first collect several ideas, lessons, and tools learned from these works that may be useful for other problems. We then briefly introduce the theory of decomposition spaces and formulate a problem for which the results of these papers may provide valuable insight.

Contents

1 Simplicial structure arises naturally in decomposition

Guideline: A decomposition naturally gives rise to a simplicial structure.

Recall the nerve of a category $\mathcal{C}$ is a simplicial set consisting of the following data:

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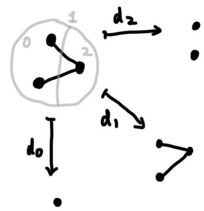

- an n-simplex is a family of n composable arrows
- The outer coface maps d_0 and d_n discard the 0th and nth arrows, respectively, while the inner coface maps compose the corresponding adjacent arrows.
- the codegeneracy map s_i is to insert identity map in the ith place.

It may be quite strange at first glance. However, it is a common and useful construction for simplicial structures.

Example 1.1. (Example 1.1.5 in [4]) Let X_k be the groupoid of graphs with an ordered partition of the vertex set into k parts, and X_0 be the empty graphs. Then the outer coface maps d_0, d_n discard the 0th and nth part, respectively, and the inner coface maps compose the corresponding adjacent arrows. The codegeneracy map inserts an empty graph at the corresponding position.

If one views rooted trees as a graph without loops, then one has a simplicial groupoid of rooted trees, see [Section 3.3, [2]]. Note that there are many variants of trees. : combinatorial trees, decorated trees, operadic trees, $\cdots$ All of them can equip a simplicial structure similar to above.

2 Always try to use morphisms to reformulate the problems

Slogan: Use the language of morphisms.

One of the spirits of category theory is to play with morphisms, rather than objects. The Yoneda Lemma can first convince it, but it is abstract. Here are some elementary and detailed examples that strengthen this idea.

All the following examples reformulate the problem in terms of morphisms, so we can use fibers, surjections, or diagrams to process them. Only when one reformulates them in the language of morphisms can one explore further and apply the general theory of decomposition space for these examples.

Example 2.1. (Remark 1.2.9, section 3.2.1 and 3.2.3 in [4])An interval $[x, y]$ of a poset P with $x, y \in P$ is defined as $[x, y] := \{z \in P | x \leqslant z \leqslant y\}$. We view P

as a category, then $[x, y] = \{$factorization of the unique morphism $x \to y$ in $P\}$. Note that d_1 means the composition of two arrows. So $[x, y]$ consists of fibers of $x \to y$ in $N(P_1)$ of $d_1 : N(P)_2 \to N(P)_1$ where $N(P)$ is the nerve of P.

Example 2.2. Given a set $\underline{n}$ with n elements and a set $\underline{m}$ with m elements. A partition of a set $\underline{n}$ into m parts is a surjection $\underline{n} \to \underline{m}$. The fiber of $\underline{n} \to \underline{m}$ illustrates the blocks that we partitioned.Let $\sigma : \underline{n} \to \underline{m}, \tau : \underline{n} \to \underline{m'}$ be two partition of $\underline{n}$ and σ is a refinement of τ. Then there exists a unique map $\underline{m} \to \underline{m'}$ such that the diagram commutes:

$$\begin{tikzcd} & \underline{n} \arrow[dl, two heads] \arrow[dr, two heads] & \\ \underline{m} \arrow[rr, dashed, "\exists!"] & & \underline{m'} \end{tikzcd}$$

Further details and applications can be seen in [Section 2.4.2,[4]].

3 On three-dimensional commutative diagrams

Guideline: Do not forget three-dimensional commutative diagrams.

Planar commutative diagrams are commonly used, whereas three-dimensional commutative diagrams are often overlooked. This series of papers provides several examples of the usage of $3D$ commutative diagrams.

Example 3.1. ([proof of Lemma 2.11, [2]]) Consider the following commutative diagram of ∞-groupoids:

$$\begin{tikzcd} Y_{n+1} \arrow[rr] \arrow[dr] \arrow[dd] & & Y_n \arrow[dr] \arrow[dd] & \\ & Y_n \arrow[rr] \arrow[dd] & & Y_{n-1} \arrow[dd] \\ X_{n+1} \arrow[rr] \arrow[dr] & & X_n \arrow[dr] & \\ & X_n \arrow[rr] & & X_{n-1} \end{tikzcd}$$

If the bottom, left, and right faces are pullbacks. With Lemma 1.11 in [2], it can be proved that the top face is a pullback. .

Other three-dimensional commutative diagrams can be found in Proposition 4.2, Lemma 6.7, and Example 6.8 in [2].

4 Some techniques that may be useful in other contexts

In this section, I collect some tools that may be useful for other problems. How they are applied is described in Section 5.

4.1 Homotopy pullback and homotopy fiber

Definition 4.1. Let $f : X \to Z$ and $g : Y \to Z$ be two maps between groupoids. Then the homotopy pullback of f and g is the groupoid $X \times_Z Y$ consisting of the following data:

- objects:$(x, y, \phi : f(x) \to g(y))$
- morphisms between (x, y, ϕ) and (x', y', ϕ') is a pair $(\alpha : x \to x', \beta : y \to y')$ such that the diagram commutes:

$$\begin{tikzcd} f(x) \arrow[r, "f(\alpha)"] \arrow[d, "\phi"'] & f(x') \arrow[d, "\phi'"] \\ g(y) \arrow[r, "g(\beta)"] & g(y') \end{tikzcd}$$

With higher morphisms, we can see things more flexibly.

The pullback in the ordinary category theory is rigid since there are no 2-morphisms that can witness isomorphism and identity. However, for groupoids, such 2-morphisms exist.

Definition 4.2. We let $\lceil b \rceil : 1 \to B$ be the map that picks the object $b \in B$. Then the homotopy fiber of $p : E \to B$ is the pullback $E_b := 1 \times_B E$.

Remark 4.3. Analog to the classical results, we can define the loop groupoid at $b \in B$ as the pullback:

$$\begin{tikzcd} \Omega_b B \arrow[r] \arrow[d] & 1 \arrow[d, "\lceil b \rceil"] \\ 1 \arrow[r, "\lceil b \rceil"'] & B \end{tikzcd}$$

4.2 Grothendieck construction and Homotopy sum

This section is a summary of section 3.6 in [1].

Main idea of Grothendieck construction: Let B be a grouoid. A family of groupoids indexed by B can be described by a functor $F : B \to \mathbf{Grpd}$ where $\mathbf{Grpd}$ is the category of groupoids. We can construct a groupoid E and a functor $F : E \to B$ such that the fiber of this functor encodes the information of "grouoids indexed by objects of B".

Conclusion 4.4. (functor of homotopy fiber) Let $p : E \to B$ be a functor between groupoids. Then we can construct a functor $E_{-} : B \to \mathbf{Grpd}$ as following:

- For objects, $b \mapsto E_b$.
- For morphism $f : b \mapsto b'$ in B, the image is a functor $E_b \to E_{b'}$ defined as following:
  - $(1, e, \phi : b \simeq p(e)) \mapsto (1, e, \phi f^{-1} : b' \to p(e))$
  - Let $(id_1, \alpha : e \to e')$ be a map between $(1, e, \phi : b \simeq p(e))$ and $(1, e', \phi' : b \simeq p(e'))$. Then the image of this morphism is $(id_1, \alpha) : (1, e, \phi f^{-1} : b' \to p(e)) \to (1, e', \phi' f^{-1} : b' \to p(e'))$

Conclusion 4.5. (Grothendieck construction for $F : B \to \mathbf{Grpd}$) For a functor $F : B \to \mathbf{Grpd}$, we have Grothendieck construction, groupoid $\int^{b \in B} F(b)$, consisting of the following data:

- objects are pairs $(b \in B, x \in F(b))$
- a morphism from (b, x) to (b', x') is a pair $(\sigma : b \to b', \phi : F(\sigma)(x) \xrightarrow{\simeq} x')$

Definition 4.6. For a functor $p : E \to B$, consider the homotopy fiber functor $E_{-} : B \to \mathbf{Grpd}$. Apply Grothendieck construction to E_{-}, we obtain $\int^{b \in B} E_b$. We call it the homotopy sum of the fibers of p.

Remark 4.7. The main idea of homotopy sum is to glue a family of groupoids indexed by groupoid B, $\{F(b) \in \mathbf{Grpd}\}_{b \in B}$ by "gluing" objects and their images. Here in the setting of groupoids, we are more flexible, and the "gluing" means there is an isomorphism between them.

For a map between sets $p : X \to Y$, we have $X = \cup_{y \in Y} p^{-1}(y)$, i.e., X is the union of fibers. Similar properties occur for homotopy sums in the following proposition.

Proposition 4.8. Let $p : E \to B$ be a functor between groupoids. Then $E \simeq \int^{b \in B} E_b$.

The following proposition shows that the homotopy sum can be written as the disjoint union of quotient groupoids (for homotopy quotient, see [Section 3.5, [1]].

Proposition 4.9. Given a functor $p : E \to B$, we have $E \simeq \sum_{b \in \pi_0(B)} E_b / Aut(b)$ where $Aut(b) = B(b, b)$.

Propositions 4.8 and 4.9 lead to the following formula representing homotopy sum as a disjoint union of quotients of groupoids.

Corollary 4.10. Given a functor $p : E \to B$, we have

$$\int^{b \in B} E_b = \sum_{[b] \in \pi_0(B)} E_b / Aut(b)$$

4.3 Trees and forests

The main idea of Section 2 is to reformulate problems with morphisms. In our first version, a tree is a graph without loops. How can we reformulate it with morphisms?

- Let N be the finite set of nodes and A be the set of edges of the tree. We suppose our tree can have only one outgoing edge per node, but can have several incoming edges.
- Since each node can have only one outedge, we use a morphism $t : N \to A$ to characterize the unique outedge of node b.
- A node b has several input edges, so we can not use a morphism to describe it directly. Let $M_b := \{(b \in N, e \in A) | e \text{ is one of the input edges of } b\}$. Let $M := \sum_{b \in N} M_b$ and $p : M \to N$, $s : M \to A$ be projections. Take fibers of p over $b \in N$ and apply $s : M \to A$, then we can obtain an input edge of b.

Therefore, we have the following definition:

Definition 4.11. A tree is a diagram of finite sets

$$A \xleftarrow{s} M \xrightarrow{p} N \xrightarrow{t} A$$

satisfying:

- t is injective
- s is injective with $A = Ims \cup 1$ where 1 is a singleton and is called root. It can also be written as $A = 1 + M$
- Define the walk-to-the-root function $\sigma : A \to A$ by $1 \mapsto 1$, $e \mapsto tp(e)$
- For any $x \in A$, there exists $k \in \mathbb{N}$ such that $\sigma^k(x) = 1$

Remark 4.12. A leaf does not come out from a node, so a leaf is the edge that does not appear in the image of t. The root edge does not have a target node, so the root is the edge that is not in the image of s.

This notation can easily describe different kinds of tree embeddings: different embeddings corresponding to different pullback conditions; see section 4.3 in [1].

There is also a valuable and elegant way to describe forests, namely as disjoint unions of trees, which will be convenient for our purposes.

Construction 4.13. Let **Bij** be the groupoid of finite sets and bijections. Let **TEmb** be the category of trees and tree embeddings. Then $\mathbf{Bij}_{/\mathbf{TEmb}}$ is the groupoid of forests. Indeed, an object in $\mathbf{Bij}_{/\mathbf{TEmb}}$ is a map between groupoids $X \to \mathbf{TEmb}$, assigning each element in the finite set X a tree, totally forming a forest.

Remark 4.14. One may be confused that **TEmb** is not an object in **Bij**, so $\mathbf{Bij}_{/\mathbf{TEmb}}$ does not make sense. Here we regard $Bij \subset$ **Grpd** and $\mathbf{Bij}_{/\mathbf{TEmb}}$ is the full subgroupoid of $\mathbf{Grpd}_{/\mathbf{TEmb}}$ consisting of objects $X \to \mathbf{TEmb}$ with $X \in \mathbf{Bij}$

We can even view trees and forests as P-trees and P-forests, which provides a functorial way to decorate each edge and node with a color (see [Section 4.4,[1]]).

5 A brief overview of decomposition spaces

5.1 Decomposition spaces versus Segal spaces

Let $\mathcal{S}$ be the category of ∞-category of ∞-groupoids. Let **Grpd** be the 2-category of 1-groupoids.

Definition 5.1. A Segal space is a simplicial space $X : \Delta \to \mathcal{S}$ fulfilling the Segal condition:

$$X_r \simeq X_1 \times_{X_0} X_1 \times \cdots \times_{X_0} X_1$$

The four equivalent definitions of Segal spaces are in [Lemma 2.10, [2]].

Definition 5.2. A decomposition space is one for which the image of any pushout in Δ of an active map along an inert map is a pullback in $\mathcal{S}$.

The definitions of active and inert maps are given in [Section 2.2, [2]].

For an equivalent definition of decomposition space, see [Proposition 3.5, [2]]
.

Remark 5.3. We need the decomposition space because its condition will make the comultiplication on $\mathbf{Grpd}/X_1$ coassociative.

Remark 5.4. In short, a Segal space can be viewed as a space built by elements in X_1 and X_0. Any r composable family of 1-simplices in X_1 can determine a unique r-simplex in X_r up to isomorphism. Roughly speaking, simplices in a Segal space can be decomposed and composed, while simplices in a decomposition space can be decomposed but may not be composed (see [Example 1.1.5, [4]]). So it leads to the following property.

Proposition 5.5. Any Segal space is a decomposition space.

Homotopy slice category $\mathbf{Grpd}_{/U}$
Let $U \in$ **Grpd**. Define $\mathbf{Grpd}_{/U}$ to be the homotopy slice category whose

- objects are functors $G \to U$ with G a groupoid,
- A morphism between $G \to U$ and $H \to U$ is a map $G \to H$ together with a natural isomorphism from $G \to U$ to $G \to H \to U$

$$\begin{tikzcd} G \arrow[rr] \arrow[dr] & & H \arrow[dl] \\ & X & \end{tikzcd}$$

a categorified analogue of a vector space

We give the following definitions on $\mathbf{Grpd}_{/U}$, which makes it behave as a categorified analogue of a vector space:

- Basis: maps $\ulcorner b \urcorner : 1 \to U$, indexed by $[b] \in \pi_0 U$. Here $\pi_0 U$ is the collection of isomorphic classes of $ob(U)$ (somewhere called connected components of U), $b \in ob(U)$, and $[b]$ is the isomorphic classes represented by b.

- Scalar multiplication: for a groupoid A,

$$A \ulcorner b \urcorner := A \to 1 \xrightarrow{\ulcorner b \urcorner} U$$

- Addition: Homotopy sums, see Section 4.2.

Proposition 5.6. For any $f: E \to U$ in $\mathbf{Grpd}_{/X}$, f can be represented as a linear combination of basis:

$$f \simeq \int^{e \in E} \ulcorner f(e) \urcorner \ \simeq \ \int_{b \in U} E_b \cdot \ulcorner b \urcorner.$$

Remark 5.7. It is a categorified analogue of a vector space, and one of the main differences from the ordinary vector space is that there is no inverse under homotopy sum. For example, $-\ulcorner b \urcorner$ does not make sense.

Coalgebra structure on $\mathbf{Grpd}_{/X_1}$ An ordinary coalgebra structure over a field is a vector space together with a compatible comultiplication and counit.

Let X be a simplicial space. The coalgebra structure on $\mathbf{Grpd}_{/X_1}$ consists of the compatible following structures:

- categorified analogue of a vector space

- comultiplication $\Delta: \mathbf{Grpd}_{/X_1} \to \mathbf{Grpd}_{/X_1 \times X_1}$,

- counit $\varepsilon: \mathbf{Grpd}_{/X_1} \to \mathbf{Grpd}_{/1}$, where 1 is the singleton set viewed as groupoid.

Definition 5.8. $(\mathbf{Grpd}_{/X_1}, \Delta, \epsilon)$ is called incidence coalgebra of the decomposition space.

Since $\mathbf{Grpd}_{/1} \simeq \mathbf{Grpd}$, the counit is a map

$$\varepsilon: \mathbf{Grpd}_{/X} \to \mathbf{Grpd}.$$

Linear functors induced by spans

A span of groupoids

$$A \xleftarrow{f} B \xrightarrow{g} C$$

induces a functor

$$g_! f^*: \mathbf{Grpd}_{/A} \to \mathbf{Grpd}_{/C}.$$

defined as $[G \to A] \mapsto [f^*G \to B \xrightarrow{g} C$ where f^*G is the pullback of $[G \to A]$ along f.

Such functors are called linear functors.

Let X be a decomposition space, i.e., a simplicial space with good conditions ensuring the comultiplication of $\mathbf{Grpd}_{/X_1}$ satisfies associativity.

Main idea: We use the face and degeneracy maps to construct a span and then induce Δ and ϵ.

The comultiplication

$$\Delta \colon \mathbf{Grpd}_{/X_1} \to \mathbf{Grpd}_{/X_1 \times X_1}$$

is induced by the span

$$X_1 \xleftarrow{d_1} X_2 \xrightarrow{(d_2, d_0)} X_1 \times X_1.$$

The counit

$$\varepsilon \colon \mathbf{Grpd}_{/X_1} \to \mathbf{Grpd}_{/1}$$

is induced by the span

$$X_1 \xleftarrow{s_0} X_0 \to 1.$$

Nerve vs. fat nerve of a category

Let $\mathcal{C}$ be a category. The nerve $N(C)$ is a simplicial set:

- 1-simplices are arrows $f \in \mathrm{Mor}(C)$,
- 2-simplices are composable pairs f_1, f_2,
- n-simplices are composable strings $f_1, \dots, f_n$.

The fat nerve of $\mathcal{C}$ is a simplicial groupoid:

- objects are strings of composable arrows,
- morphisms are levelwise isomorphisms between such strings, rendering the diagram commutes:

$$\begin{tikzcd}
\bullet \arrow[r] \arrow[d, "\cong"'] & \bullet \arrow[r] \arrow[d, "\cong"'] & \bullet \arrow[r, dotted, no head] \arrow[d, "\cong"'] & \bullet \arrow[r] \arrow[d, "\cong"'] & \bullet \arrow[d, "\cong"'] \\
\bullet \arrow[r] & \bullet \arrow[r] & \bullet \arrow[r, dotted, no head] & \bullet \arrow[r] & \bullet
\end{tikzcd}$$

Hanger category

The hanger category has objects and arrows

$$\begin{tikzcd}
 & \bullet \arrow[loop above, "j"] \arrow[dr, "b"] & \\
d_1 f \arrow[ur, "a"] \arrow[rr, "f"'] & & d_0 f
\end{tikzcd}$$

$$ab = f, \quad jj = 1, \quad aj = a, \quad jb = b.$$

Explicit computation for the hanger category

The comultiplication$\Delta\colon \mathbf{Grpd}_{/X_1} \to \mathbf{Grpd}_{/X_1 \times X_1}$ maps $\ulcorner f \urcorner : 1 \to X_1$ to $1 \times_{X_1} X_2 \to X_2 \xrightarrow{(d_2,d_0)} X_1 \times X_1$ where the first map comes from pullback of $\ulcorner f \urcorner$ along $d_1 : X_2 \to X_1$.

$ob(1 \times_{X_1} X_2) = \{(*, \sigma, \phi : f \simeq d_1\sigma) | \sigma \in X_2, * \in 1\}$

- $\sigma \in X_2$, so σ is of the form $f_1 \circ f_2$ with $f_1, f_2 \in \{a, b, f, j, \mathrm{id}\}$.Since $d_1\sigma \simeq f$ (the composition of σ are isomorphic to f), the possibility of σ are $\xrightarrow{f}\xrightarrow{id}$, $\xrightarrow{id}\xrightarrow{f}$, $\xrightarrow{a}\xrightarrow{b}$
- Since $\mathrm{Hom}(d_0f, d_0f) = \{id\}$, $\mathrm{Hom}(d_1f, d_1f) = \{id\}$, ϕ in above triple could only be id.

Hence, $Ob(1 \times_{X_1} X_2) = \{(*, \xrightarrow{f}\xrightarrow{id}, id_f),\ (*, \xrightarrow{id}\xrightarrow{f}, id_f),\ (*, \xrightarrow{a}\xrightarrow{b}, id_f)\}$.

$$\begin{aligned}
\Delta(f) &= 1 \times_{X_1} X_2 \xrightarrow{\text{projection}} X_2 \xrightarrow{(d_2,d_0)} X_1 \times X_1 \\
&= \int^{(*,\sigma,\mathrm{id}) \in 1 \times_{X_1} X_2} \ulcorner (d_2\sigma,\ d_0\sigma) \urcorner \\
&= \int^{(*,\sigma,\mathrm{id}) \in 1 \times_{X_1} X_2} \ulcorner d_2\sigma \urcorner \otimes \ulcorner d_0\sigma \urcorner \\
&= \sum_{[(*,\sigma,\mathrm{id})] \in \pi_0(1 \times_{X_1} X_2)} |\mathrm{Aut}(*, \sigma, \mathrm{id})|^{-1}\ \ulcorner d_2\sigma \urcorner \otimes \ulcorner d_0\sigma \urcorner
\end{aligned}$$

Then let's compute $\pi_0(1 \times_{X_1} X_2)$. Suppose there is a map between $(*, idf, id)$ and $(*, fid, id)$:

$$\begin{array}{ccccc}
d_1f & \xrightarrow[f]{} & d_0f & \xrightarrow[id]{} & d_0f \\
\downarrow & & \downarrow & & \downarrow \\
d_1f & \xrightarrow{id} & d_1f & \xrightarrow{f} & d_0f
\end{array}$$

The middle verticle isomorphism does not exist, since $\mathrm{Hom}(d_0f, d_1f) = \emptyset$
For the same reason one can show $\pi_0(1 \times_{X_1} X_2) = ob(1 \times_{X_1} X_2)$

$$\begin{aligned}
\Delta(f) = &\left|\pi_1\big(1 \times_{X_1} X_2,\ (*,\ fid, id)\big)\right|^{-1} \ulcorner f \urcorner \otimes \ulcorner id \urcorner \\
&+ \left|\pi_1\big(1 \times_{X_1} X_2, (*, idf,\ id)\big)\right|^{-1}\ \ulcorner id \urcorner \otimes \ulcorner f \urcorner \\
&+ \left|\pi_1\big(1 \times_{X_1} X_2,\ (*,\ ab,\ id)\big)\right|^{-1}\ \ulcorner a \urcorner \otimes \ulcorner b \urcorner \\
&= \ulcorner f \urcorner \otimes \ulcorner id \urcorner\ +\ \ulcorner id \urcorner \otimes \ulcorner f \urcorner\ +\ \frac{1}{2} \ulcorner a \urcorner \otimes \ulcorner b \urcorner
\end{aligned}$$

Here $\frac{1}{2}$ comes from the fact that$\mathrm{Hom}_{1\times_{X_1}X_2}\big(ab,\,ab\big)$ consists of two morphisms:.

$$(id, j, id) \quad \text{and} \quad (id, id, id).$$

5.2 Möbius inversion

Classical Möbius inversion on $\mathrm{Map}(\mathbb{N}^{\times}, \mathbb{C})$

Consider $\mathrm{Map}(\mathbb{N}^{\times}, \mathbb{C})$ be the set of maps from $\mathbb{N}^{\times}$ to $\mathbb{C}$. To introduce Möbius inversion, we need the following data:

- Define multiplication on $\mathrm{Map}(\mathbb{N}^{\times}, \mathbb{C})$: $f * g(n) := \sum\limits_{i \cdot j = n} f(i)g(j)$, called convolution
- unit of this multiplication $\epsilon : \mathbb{N}^{\times} \to \mathbb{C}$ is defined by $\epsilon\,(n) = \begin{cases} 1 & n = 1 \\ 0 & otherwise \end{cases}$ $(f * \epsilon(n) = f(n) \cdot \epsilon(1) = f(n))$
- Define the function $\zeta : \mathbb{N}^{\times} \to \mathbb{C}$ to be the constant function equal to 1.
- Define Möbius function

$$\mu(n) = \begin{cases} 0, \text{if } n\text{contains a square factor} \\ (-1)^r \text{if } n \text{ is a product of } r \text{ primes} \end{cases}$$

Möbius inversion principal means, for any $f, g \in Map(\mathbb{N}^{\times}, \mathbb{C})$,

$$\text{If } f = g * \zeta, \text{ then } g = f * \mu$$

Remark 5.9. Apply Möbius inversion principal to $\zeta = \epsilon * \zeta$, we have $\epsilon = \zeta * \mu$. The convolution on $Map(\mathbb{N}^{\times}, \mathbb{C})$ is commutative, so we also have $\mu * \zeta = \epsilon$. So ζ is the convolution inverse of μ.

Generalized Möbius inversion

Let (C, Δ, ϵ) be a coalgebra and (A, m, u) be an algebra. Let $Lin(C, A)$ be the space of linear maps. Let $\Delta(x) = \sum_{(x)} x_{(1)} \otimes x_{(2)}$ for $x \in C$. Then we define convolution in $Lin(C, A)$ as $\alpha * \beta(x) = \sum_{(x)} m(\alpha(x_{(1)}), \beta(x_{(2)}))$.

Proposition 5.10. Define $e = u\epsilon$. Then $(Lin(C, A), - * -, e)$ form an algebra.

Remark 5.11. In this case, the convolution may not be commutative.

Question: For a $\phi \in Lin(C, A)$, when does ϕ has inverse and how to compute its inverse? A special case is solved in the following theorem.

Theorem 5.12 (Theorem 4.3 in [5])**.** If $\phi \in Lin(C, A)$ sends all group-like elements (i.e., element x satisfying $\Delta(x) = x \otimes x$) to 1, then ϕ has inverse ψ fulfilling $\psi = e - \psi * (\phi - e)$.

An iterative, recursive construction and an alternating sum yield the inverse ψ. Let $\psi_0 = e$, $\psi_{n+1} = \psi_n * (\phi - e)$. Let $\psi_{even} = \sum_{n \quad even} \psi_n$ and $\psi_{odd} = \sum_{n \quad odd} \psi_n$. Then we let $\psi = \psi_{even} - \psi_{odd}$, and we can prove that this ψ satisfies the required recursive relation.

5.3 How do we comprehend negative sets?

This remains an open question arising naturally in the study of generalized Möbius inversion. At the current level of understanding, some aspects of this problem cannot yet be formulated clearly.

Let X be a decomposition space. Then $(\mathbf{Grpd}_{/X_1}, \Delta, \epsilon)$ forms an incidence coalgebra. It is therefore natural to ask whether there exists an analogue of the Möbius inversion theorem for elements of $\mathrm{Fun}(\mathbf{Grpd}_{/X}, \mathbf{Grpd})$, analogous to Theorem 5.12.

Analogous to the construction in the proof of Theorem 5.12, Imma Gálvez-Carrillo, Joachim Kock, and Andrew Tonks construct functors $\Phi_{even}, \Phi_{odd} \colon \mathbf{Grpd}_{/X_1} \to \mathbf{Grpd}$. However, compared with the expression $\psi_{even} - \psi_{odd}$, the meaning of "$\Phi_{even} - \Phi_{odd}$" is unclear.The expression "$\Phi_{even} - \Phi_{odd}$" would become meaningful once we understand what it means to take the difference of two sets, and consequently, of two groupoids.

This leads to a fundamental question: by analogy with negative numbers, what should be meant by negative sets?

The following remark was pointed out by Professor Joachim Kock: When A and B are sets, the expression "$A-B$" should *not* be understood as cancellation by elements. When A and B are infinite countable sets, then there is a bijection between A and B, and then A and B will cancel each other, which is not the result we want. Thus, cancellation should be encoded by additional structure rather than by naive set-theoretic subtraction.

Luckily, when we work with complete decomposition spaces, we can clearly understand this problem.

Theorem 5.13 (Theorem 3.8, [3])**.** For a complete decomposition space, the following Möbius principle holds:

$$\zeta * \Phi_{even} = \varepsilon + \zeta * \Phi_{odd}$$

$$\Phi_{even} * \zeta = \varepsilon + \Phi_{odd} * \zeta$$

Complete decomposition spaces provide a natural setting for Möbius inversion at the objective level. For more details, see [Section 3.3, [4]].

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